

AUTOKART RC

Development, Testing & Actuator Control Report

30 July 2026 - 02 September 2026

Purpose

Consolidated engineering documentation of AutoKart RC architecture decisions, stepper and BLDC bench testing, RC/full-kart trials, fault diagnosis, and the current actuator-control implementation.

Steering

NEMA 23 + TB6600
Open-loop STEP/DIR/ENA

Brake

NEMA 23 + TB6600
0 to +15 deg software range

Traction

BLD750 + BLDC
Analog speed + digital EN/BRK/F/R

Control

ESP32-S3 bench controller
Browser UI + safety state machine

Executive Summary

Engineering status as of 02 September 2026

The AutoKart RC development progressed from isolated stepper testing to a proposed vehicle-control architecture, BLDC bench validation, RC operation, and full-kart testing. The major unresolved issue is steering torque under road load: steering behaves correctly while the kart is jacked, but the stepper stalls or vibrates when the tyres are loaded.

The latest mechanical check changed the assumed steering reduction from 3:1 to approximately 2:1. Using the working motor-torque assumption of 3 N·m, the ideal geared output is therefore about 6 N·m. Applying the same loss range used in the earlier estimate gives roughly 4.7-5.1 N·m practical output; this remains an engineering estimate and should be validated experimentally.

Current control milestone

BLD750 digital control has been added using PN2222A transistor interfaces: GPIO13 -> EN, GPIO14 -> BRK, and GPIO15 -> F/R. The attached implementation source defines transistor-on as pulling the BLD750 input low. Direction reversal is allowed only from a safe stopped state.

Key engineering findings

- Stepper steering motion can achieve the commanded 90° movement under no-load/jacked conditions.
- Full-kart steering fails under tyre/road load by stepping, vibrating, or failing to rotate the steering gear.
- BLD750 fault code sequence encountered during garage testing: six red blinks (Hall sensor error), followed later by seven red blinks (rotor locked).
- The Hall-sensor-related fault was addressed through U/V/W wiring correction; the rotor-lock condition was cleared after adjustment of the potentiometer anti-clockwise to its click position, according to the test log.
- The integrated controller remains open-loop for steering and brake; commanded angle is inferred from pulse count and assumed physical zero, with no encoder/limit-switch feedback in the documented implementation.

Current actuator signal map

Function	ESP32-S3	Interface	Current role
Steering	GPIO4 / 5 / 6	TB6600 STEP / DIR / ENA	Open-loop steering position
Brake	GPIO10 / 11 / 12	TB6600 STEP / DIR / ENA	0 to +15° application
Throttle	GPIO8 SDA / GPIO9 SCL	GP8630N -> BLD750 SV	Analog speed request
BLD750 Enable	GPIO13	PN2222A -> EN	RUN / STOP
BLD750 Brake	GPIO14	PN2222A -> BRK	Brake released / quick brake
BLD750 Direction	GPIO15	PN2222A -> F/R	CW / CCW selection

Development Timeline

Chronological engineering record

30/07/2026	Stepper motor testing 90° steering-turn test controlled through ESP32. This established basic stepper command generation and direction control before full vehicle integration.
08/08/2026	AutoKart architecture proposed Vehicle Control Unit (VCU) concept proposed. E-stop concept introduced. LoRa RA-02 selected for the emergency-stop communication path. Initial bill of materials (BOM) proposed.
19/08/2026	BLDC test bench - analog speed path Bench chain recorded as 24 V -> BLD750 -> BLDC, with BLD750 speed command provided through a GP8630N converter controlled by ESP32 and commanded from a laptop.
22/08/2026	BLDC test bench - speed-range test 24 V -> BLD750 -> BLDC with 0-5 V converter -> ESP32 control path. High-to-low RPM testing conducted. Garage assembly also tested BLDC and steering without the final mount.
22/08/2026	BLD750 fault diagnosis Six red blinks observed and recorded as Hall sensor error; resolved by correcting U/V/W wiring. Seven red blinks then observed and recorded as rotor locked; resolved by turning the potentiometer anti-clockwise to the click position.
25/08/2026	RC testing Steering tested through 90° range. Brake command configured clockwise with a maximum 15° application.
28/08/2026	Full kart test Steering was satisfactory while the vehicle was jacked. Under road/tyre load, the stepper could not drive the steering mechanism reliably and instead stepped or vibrated.
30/08/2026	Stepper torque review Working conclusion: NEMA 23 motor treated as approximately 3 A and 3 N·m. With the then-assumed 3:1 reduction, ideal output was estimated at 9 N·m and practical output at roughly 7-8 N·m after mechanical losses.

02/09/2026

Gear-ratio correction and BLD750 digital control

Measured/identified steering ratio revised to approximately 1:2 rather than 1:3. Ideal output torque becomes about 6 N·m for a 3 N·m motor; practical torque requires validation. BLD750 EN/BRK/F/R control completed using PN2222A transistor interfaces on GPIO13/14/15.

System Architecture & Actuator Control

Current documented bench/integration implementation

Control flow

Command path

Browser UI -> ESP32-S3 HTTP server/state machine -> TB6600 stepper interfaces and GP8630N/BLD750 traction interface.

Steering

- NEMA 23 stepper motor driven through TB6600.
- GPIO4: STEP/PUL, GPIO5: DIR, GPIO6: ENA.
- Software range in the attached implementation: -90° to +90°.
- Position is open-loop; angle is calculated from commanded pulse count rather than measured shaft/road-wheel position.

Brake

- Second NEMA 23/TB6600 axis.
- GPIO10: STEP, GPIO11: DIR, GPIO12: ENA.
- Positive motion is clockwise brake application.
- Documented software range: 0° to +15°.

Throttle / traction

- ESP32 communicates with the GP8630N over I²C using GPIO8 SDA and GPIO9 SCL.
- GP8630N OUT feeds BLD750 SV, with common ground to BLD750 COM.
- The attached v5 implementation clamps commanded throttle to 0-3.0 V even though the DAC is configured for a 0-10 V output range.
- Throttle is accepted only while armed and with the brake released; the controller ramps output rather than applying an instantaneous step.

BLD750 digital input control via PN2222A

GPIO	BLD750 input	Transistor action	Functional interpretation
13	EN	GPIO high -> transistor ON -> input pulled low	RUN; GPIO low/open -> STOP
14	BRK	GPIO high -> transistor ON -> input pulled low	Brake released; low/open -> quick brake
15	F/R	GPIO state selects transistor/input condition	Direction selection (software labels CW/CCW)

Safe-direction rule

The standalone BLD750 test logic permits direction reversal only after the drive reaches the full safe stopped state: DAC command at 0 V, BLD750 brake applied, and enable stopped/disabled.

Faults, Diagnostics & Resolutions

Observed issues during garage and vehicle testing

Date	Subsystem	Symptom / indication	Action taken	Status
22/08	BLD750 / BLDC	6 red blinks; logged as Hall sensor error	Corrected U/V/W wiring	Resolved in test log
22/08	BLD750 / BLDC	7 red blinks; logged as rotor locked	Potentiometer turned anti-clockwise to click position	Resolved in test log
28/08	Steering	Works while jacked; steps/vibrates under road load	Torque and gearing investigated	Open
02/09	Steering gearing	Previously assumed 3:1 ratio found to be about 2:1	Recalculate available output torque; further mechanical validation required	Open / under review

Steering torque assessment

Working motor assumption: 3 N·m at the motor shaft.

Case	Ideal geared torque	Practical interpretation
Earlier 3:1 assumption	$3 \times 3 = 9 \text{ N}\cdot\text{m}$	Earlier project estimate: ~7-8 N·m after losses
Current ~2:1 ratio	$3 \times 2 = 6 \text{ N}\cdot\text{m}$	Lower than previously assumed
Estimated with same loss range	~4.7-5.1 N·m	Estimate only; not a measured steering torque

Interpretation

The ratio correction materially reduces available steering torque. Because the mechanism already fails under tyre load, the next engineering decision should be based on measured steering torque at the steering shaft/rack, driver current settings, supply voltage, step rate, and mechanical friction - not the motor rating alone.

Open-loop control limitation

The attached actuator-control implementation explicitly states that steering and brake positions are open-loop: the software tracks commanded pulses from an assumed physical zero. There is no encoder or limit-switch feedback in the documented controller. As a result, a stall or skipped step can make the software position diverge from the true mechanical position.

Recommended next validation steps

- Measure the actual steering reduction precisely using tooth counts or input/output angle measurements.
- Measure required steering torque with the kart on the ground, ideally at stationary tyre load and during slow rolling motion.
- Confirm TB6600 current and microstep switch configuration against the NEMA 23 motor rating and the pulse calibration used by software.
- Record battery/driver voltage at the TB6600 during an attempted loaded steering manoeuvre to detect voltage sag.
- Validate mechanical friction, alignment, bearing preload, steering-column binding, and tyre scrub contribution.
- Integrate encoder feedback (AS5600/AS5047P as applicable) before treating commanded steering angle as safety-critical ground truth.
- Repeat BLDC EN/BRK/F/R testing at low speed with a clear safe-stop and direction-reversal procedure before road testing.

Safety & State-Machine Logic

Extracted from the attached actuator-control documentation

Event	Documented controller response
Boot	Throttle 0 V; steering center; brake applied; controller remains disarmed.

Arm	Throttle remains 0; steering holds; brake releases; commands may then be accepted.
Brake ON	Throttle is zeroed immediately; brake target moves to +15° clockwise.
Brake OFF	Brake returns to 0°; throttle remains forced to zero until a new throttle command is issued.
Controlled stop	Controller disarms, ramps throttle down, then applies brake once throttle is near zero.
Heartbeat timeout	If no heartbeat is received for 1.8 s while armed: disarm, throttle zero, steering hold, brake apply.
E-stop	Latched state: throttle zero, steering hold, brake apply, commands blocked until reset; reset leaves system disarmed.

Bench-test warning from source

The attached source notes a previous BLD750 fault near 0.75 V and states that the 3.0 V software ceiling was not electrically validated at the time of that report. Treat the traction command range as bench-test configuration until verified on the actual hardware.

Source notes & terminology

- Primary source attachment: “AutoKart RC - Actuator Control Report”, dated 02 September 2026.
- Primary integrated software identified by the source: `autokart_rc_v5_controlled_stop.py`.
- Standalone BLD750 digital-input test identified by the source: `full_blcdc_test.py`.
- The 19 August BLDC test-bench entry uses the GP8630N converter.
- Torque figures in this report are engineering calculations from the project’s stated 3 N·m working assumption and gear-ratio estimates; they are not dynamometer measurements.

Project status - 02 September 2026

Area	Status	Next gate
Stepper control	Functional under no-load	Validate loaded torque and feedback
Brake motion	Bench/RC logic established	Mechanical validation under real braking load
BLDC speed control	Bench-tested	Validate command scaling and repeat safe low-speed test
BLD750 EN/BRK/F/R	Control logic completed	Integrated safety validation
Full-kart steering	Not yet acceptable under load	Resolve torque/gearing/friction limitation
Architecture / VCU / E-stop	Proposed	Integrate and validate end-to-end